An RF-Energy-Harvesting Architecture for Smart Waste Monitoring: Design and Feasibility Model

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***Abstract*—Battery-powered sensing nodes can simplify waste-bin monitoring, but fleet-wide battery replacement creates recurring maintenance and material costs. This paper presents a design concept for a battery-free waste-monitoring node that combines ambient radio-frequency (RF) energy harvesting, threshold-controlled energy storage, fill-level sensing, and event-driven IEEE 802.15.4 telemetry. A mains-powered gateway associates received reports with a known collection area and forwards them to a cloud service for dispatch planning. We formulate a first-order energy-balance model and calculate idealized recharge intervals for a 5-mJ sensing-and-transmission event over a range of assumed net harvested powers. These calculations are not field measurements: ambient RF availability, end-to-end conversion efficiency, sensor and radio energy, and link reliability must be measured at deployment sites. The design therefore provides a feasibility framework and identifies the measurements required before a battery-free implementation or municipal cost benefit can be claimed.**

***Index Terms*—ambient RF energy harvesting, battery-free sensor, Internet of Things, smart waste management, IEEE 802.15.4**

I. Introduction

Municipal waste collection depends on timely information about bin fill levels and on effective dispatch planning. Internet of Things (IoT) systems can provide that information, although sensor deployment at large scale also creates maintenance and connectivity requirements. Prior work surveys smart waste-management systems, including bin sensing, data services, and collection-route planning [4].

Battery replacement is a particular concern when nodes are numerous or difficult to access. Energy harvesting may reduce battery dependence, but the usable energy depends on the source, site, antenna, power-conversion circuit, and duty cycle. Ambient RF is attractive where illumination is poor, yet measurements show that it is spatially variable and must be evaluated at the intended installation locations [1]. RF harvesting should therefore be treated as a site-dependent supply, not as guaranteed continuous power.

This paper describes a proposed architecture and a transparent first-order energy model. It does not report a fabricated prototype, field trial, or measured cost-benefit study. The contributions are: (1) an event-driven sensor and gateway architecture; (2) a sensor-validation policy intended to reduce unnecessary reports; and (3) an energy-recharge calculation that exposes the assumptions needed to assess feasibility.

II. Proposed System Architecture

The system has three tiers: a battery-free sensor node, a powered gateway, and a cloud service. The node harvests ambient RF energy and stores it in a capacitor. When the storage voltage crosses an upper threshold, a power-management circuit enables the sensing and communication subsystems. After a report or failed validation, the node returns to a low-power state. Hysteretic thresholds can prevent rapid on/off cycling, but the chosen thresholds and storage capacity must be matched to the load profile and harvester.

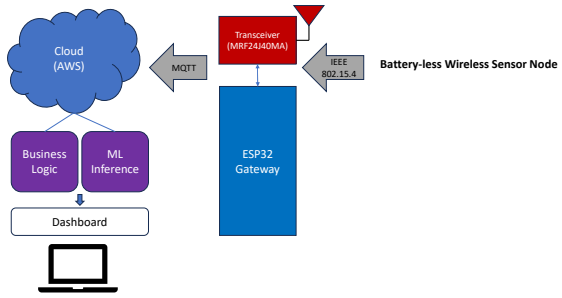


Fig. 1. End-to-end architecture showing the RF-harvesting node, powered gateway, and cloud service.

A. Sensor Node and Validation

An ultrasonic range sensor estimates the distance from the bin opening to the waste surface. A mechanical lid switch can provide an additional state input. The node reports only after applying a defined validation rule, for example requiring the lid to be closed and the fill estimate to exceed a threshold. This rule can reject some transient readings, but a lid-state switch alone does not establish that an ultrasonic reading is correct. Sensor placement, calibration, contamination, irregular waste surfaces, and repeated-sample filtering must be evaluated experimentally.

The component examples in the original design concept are a PIC24F-family microcontroller and Microchip MRF24J40MA radio module. The latter is a 2.4-GHz IEEE 802.15.4 module with published transmit and receive currents of 23 and 19 mA, respectively [2]. These currents make radio-on time and supply voltage material to the energy budget. A JSN-SR04T-class ultrasonic module is not assumed to be compatible with a microjoule budget without measurement; its operating voltage, active current, and startup behavior must be included in the design calculation.

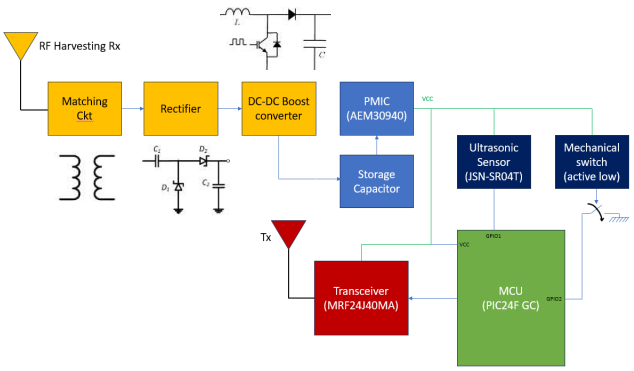


Fig. 2. Proposed node hardware, including RF harvesting, energy storage, power management, sensors, and radio.

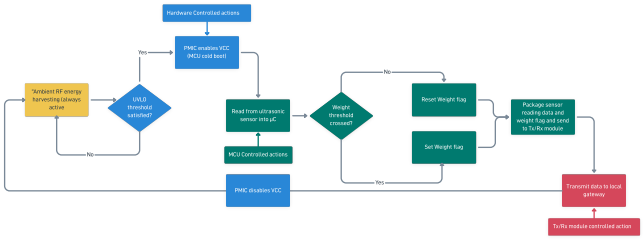


Fig. 3. Proposed sensing and validation sequence before transmission.

B. Gateway and Location Association

The node sends a short identifier and sensing result over IEEE 802.15.4 [3]. A powered gateway can add its own identifier, location, and receipt timestamp before forwarding the message to a cloud service. Gateway-based location identifies the receiving gateway or coverage area; it does not necessarily determine the exact bin location, particularly where gateway coverage overlaps or a node can reach multiple gateways. Installation records or a denser gateway layout are needed when collection requires bin-level coordinates. MQTT may be used between gateway and cloud, subject to the deployment's security and connectivity requirements.

III. Energy Model and Dispatch Logic

A. Recharge-Interval Estimate

Let E_{cycle} denote the energy consumed by one complete wake, sensing, processing, and transmission event, and let P_{net} be the average DC power available to storage after RF-to-DC conversion and power-management losses. An idealized lower-bound recharge interval is

$$T_{\text{recharge}} = \frac{E_{\text{cycle}}}{P_{\text{net}}}. \tag{1}$$

For illustration, we use $E_{\text{cycle}} = 5\,\text{mJ}$ as an unverified design assumption. The values of P_{net} in Table I are scenarios, not measured ambient-power results. The table assumes a continuous, constant net DC input and excludes leakage, capacitor limits, load variation, and missed or repeated transmissions; actual intervals can be longer.

TABLE I
Idealized recharge interval for a 5 mJ event

Assumed P_{net}	Recharge interval	Illustrative context
5 µW	16.7 min	Higher input
1 µW	1.39 h	Intermediate input
0.1 µW	13.9 h	Low input

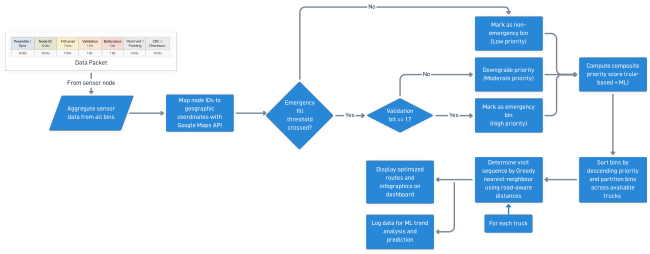


Fig. 4. Conceptual route-planning illustration for validated collection requests on a municipal road network.

The example demonstrates the direct dependence of reporting frequency on net harvested power; it does not establish that the stated power levels are available at a particular bin. Ambient-RF surveys provide useful methodology and context, but their results cannot be transferred directly to a different city, antenna, enclosure, or receiver [1]. A validation study should measure incident RF power and rectified DC power over representative sites and times, then measure the complete node's energy per event.

B. Cloud Dispatch

The cloud service can apply a fill threshold and data-quality checks before adding bins to a collection manifest. Routing is a vehicle-routing problem constrained by road topology, vehicle capacity, depot location, and service requirements. A greedy nearest-neighbor heuristic is one possible baseline, but it does not generally guarantee a globally shortest route. Any claimed distance or fuel savings should be evaluated against a stated baseline using the same road graph, demand data, and vehicle constraints. Smart waste-management research discusses route planning as part of the broader system problem [4].

IV. Illustrative Cost Scenario and Limitations

For a transparent arithmetic example, assume 50 nodes in each of 150 wards, 150 gateways, a node hardware cost of INR 1,500, and a gateway cost of INR 1,200. The resulting quantities are 7,500 nodes and 150 gateways. Hardware-only capital cost is

$$7{,}500(1{,}500) + 150(1{,}200) = \text{INR } 11{,}430{,}000, \tag{2}$$

or INR 1.143 crore (INR 76,200 per ward). The 150-ward administrative count is documented by GHMC [5]; the bin density and unit prices above are scenario assumptions, not sourced procurement data. The estimate excludes installation, enclosure, antennas, storage, maintenance, gateway backhaul, cloud service, taxes, and replacement. It therefore cannot

support a claim of economic viability or savings without quotations and a lifecycle comparison.

The principal technical limitation is the absence of measured results. In particular, feasibility depends on available RF power at the antenna, conversion efficiency at the measured input level, storage leakage, sensor energy, radio airtime and retries, and the required reporting latency. Metal bins and other obstructions may substantially change RF coupling. The architecture also requires an adequate gateway link and a fall-back strategy for periods when harvested energy is insufficient. A prototype and site survey are necessary to establish these quantities.

V. Conclusion

This paper has presented a design concept for RF-energy-harvesting waste-bin monitoring with event-driven sensing, IEEE 802.15.4 telemetry, powered gateways, and cloud-based dispatch planning. The energy model gives idealized recharge intervals for explicitly assumed net input powers and an assumed event energy; it is not a field validation. The cost calculation is likewise an illustrative hardware-only scenario. Prototype measurements, site-specific RF surveys, and route and lifecycle-cost comparisons are required before claiming reliable battery-free operation or municipal savings.

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